

Technical Note: MPI Risk Engine Refit, Bayes/Cox Integration, and EV/Risk Outputs

1. Purpose

This note documents updates to the MPI Risk Engine and associated analyses. The goal is to ensure auditability, traceability, and replicability of changes applied to the Bayesian and Cox models, their integration, and impacts on expected value (EV) and risk outputs.

2. Methods

2.1 Bayes Model

- Original: included province, sector, and province–sector interaction LRs directly, sometimes swamping smaller multipliers.
- Update: added a tunable shrinkage parameter (α) applied to `prov_sec_*` terms (default $\alpha=0.6$).

2.2 Cox Model

- Original: covariates included `cleantech_flag`, `cost_quintile`, `province`, `sector`.
- Update: added `Greenfield_flag` and `FOAK_flag`, switched to `cost_percentile`, and validated with time-split (train ≤ 2019 , test > 2019).
- Expanded coefficient file includes all provinces, sectors, cleantech categories with reference values for audit clarity.

2.3 Blending

- Original: fixed at 40/60 Bayes/Cox.
- Update: blend ratio exposed as `BLEND_BAYES_W`. Tested 20/80 to 80/20.

2.4 Validation Metrics

- AUC/C-index for discrimination.
- Brier score for calibration.
- `EV_cum`, `EV_disc`, and `k*` for expected values and risk-adjusted returns.

3. Results

3.1 Bayes Shrinkage

- $\alpha=1.0$: AUC ~ 0.863 , Brier ~ 0.189 .
- $\alpha=0.6$: AUC ~ 0.857 , Brier ~ 0.180 .
- Shrinkage improved calibration while preserving discrimination.

3.2 Cox Re-fit

- Original C-index ~ 0.72 .
- Re-fit C-index ~ 0.82 (full), ~ 0.84 (test split).
- Significant: Province BC (***) , FOAK (*) , Quebec/Other (*) .

3.3 Blend

- Bayes ($\alpha=0.6$): AUC 0.857, Brier 0.180.
- Cox re-fit: AUC 0.780, Brier 0.180.

- 50/50 blend: AUC 0.836, Brier 0.145 (best calibration).
- 60/40: AUC 0.842, Brier 0.147 (better ranking).

3.4 Project Impacts

- Downward: Darlington SMR, Red Chris, LNG Canada Phase 2, McIlvenna Bay.
- EV_cum largest declines: NWMO DGR (−4.4B), Darlington (−3.6B), Pathways CCS (−2.7B), Kitimat Clean (−1.5B).
- EV_cum increases: Nova Scotia Power Battery (+29M), Mass Timber (+25M), District Energy (+24M).

4. Implementation

- Patched engine (mpi_risk_engine_v03_patched.py) created with PROVSEC_ALPHA and BLEND_BAYES_W knobs.
- Cox coefficients expanded for audit transparency.
- Outputs validated against old v03: blended_prob ↓0.07 avg; EV_cum ↓111 avg/project; k* adjusted ↑20 points on avg.

5. Conclusions

- Refit improves discrimination and calibration.
- Bayes shrinkage prevents swamping.
- Cox now aligned with Bayes (Greenfield/FOAK, cost).
- Blends: 50/50 optimal for calibration; 60/40 better for ranking.
- EV redistributed away from mega-projects to smaller modular projects.

6. Replication Checklist

1. Use mpi_risk_engine_v03_patched.py with PROVSEC_ALPHA=0.6, BLEND_BAYES_W=0.5.
2. Point Cox to cox_refit_coefficients_timesplit_expanded.csv.
3. Train/test split (≤ 2019 vs > 2019) for validation.
4. Evaluate AUC, Brier, EV_cum, and k*.